

Admissibility Forces Hilbert Geometry and Schrödinger-Form Dynamics

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Abstract

This maximal consolidated manuscript presents a two-layer closure argument.

Layer 0 (Meta-closure): under admissibility (Standing Conservation, Impossibility Suite, ZPA) and compositional richness (symmetric monoidal independence, mixing richness), predictive geometry is *Hilbert-or-denied*: any non-Hilbert geometry induces a carrier, requires free tuning, or violates standing invariance.

Layer 1 (Geometry $\rightarrow$ dynamics): given Hilbertian geometry forced by Layer 0, a continuous reversible admissible update law is necessarily unitary and therefore has Schrödinger form.

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1 Meta-Closure: Hilbert-or-Denial

1.1 Primitives

Definition 1 (Standing Conservation). *No admissible update may create, amplify, transfer, or recover standing absent admissible witness exchange.*

Definition 2 (Impossibility Suite). *No Generators. No Carriers. No Repairs.*

Definition 3 (Zero-Parameter Admissibility (ZPA)). *Any free continuous tuning not invariance-forced fails closed.*

1.2 Categorical compositional setting

Definition 4 (System category). *Let $\mathbf{Sys}$ be a symmetric monoidal category with tensor $\otimes$ modeling independent composition.*

Definition 5 (Predictive functor). *$\mathcal{P} : \mathbf{Sys} \rightarrow \mathbf{Top}$ assigns each system A a topological predictive state space $\mathcal{P}(A)$.*

Axiom 1 (Compositional closure). *There exist continuous natural maps $\boxtimes_{A,B} : \mathcal{P}(A) \times \mathcal{P}(B) \rightarrow \mathcal{P}(A \otimes B)$ compatible with associativity/unit/symmetry.*

Axiom 2 (Mixing richness). *Each $\mathcal{P}(A)$ contains nontrivial continuous families of states and admits continuous reparameterizations that do not select privileged decompositions.*

Definition 6 (Carrier). *A carrier occurs if independent preparation in B modulates the restricted predictive content on A after an admissible update on $A \otimes B$, absent any admissible witness morphism $B \rightarrow A$.*

Definition 7 (Standing weight). *A standing weight is a family $w_A : \mathcal{P}(A) \rightarrow \mathbb{R}_{>0}$ continuous and invariant under admissible updates, multiplicative under $\boxtimes$.*

Theorem 1 (Hilbert-or-Denial). *Under Standing Conservation, the Impossibility Suite, ZPA, and Axioms 1–2, admissible predictive geometry is Hilbertian or denied (carrier/parameter/standing failure).*

Proof. See companion meta-proof for full exhaustion. Here we use it as a forced closure classification. $\square$

2 Geometry Layer: Inner-Product Reconstruction

In the Hilbertian regime, representatives exist as a complex vector space $\mathcal{V}$, rays represent states, and standing weight is quadratic. We now derive the inner product explicitly from standing weight, matching the companion reconstruction proof.

2.1 Standing weight axioms

Definition 8 (Standing weight on representatives). *$w : \mathcal{V} \setminus \{0\} \rightarrow \mathbb{R}_{>0}$ is continuous, ray-invariant, and multiplicative under independent tensor composition.*

2.2 Homogeneity

Lemma 1. *There exists α such that $w(\lambda\psi) = \lambda^\alpha w(\psi)$.*

Proof. As in the reconstruction paper: continuity + multiplicativity over positive scalars implies a power law. $\square$

2.3 Exponent forcing and quadraticity

Axiom 3 (2D isotropy / no privileged decomposition). *Every real 2D plane admits continuous isotropic mixing action consistent with admissibility.*

Theorem 2. *$\alpha = 2$, hence w is quadratic on every 2D plane.*

Proof. Coordinate derivation as in maximal hardening: isotropy forces $w(\psi) = c(x^2 + y^2)^{\alpha/2}$ on planes; quadratic representability forces $\alpha = 2$. $\square$

Define $\|\psi\| := \sqrt{w(\psi)}$.

Theorem 3 (Parallelogram identity). $\|\psi + \phi\|^2 + \|\psi - \phi\|^2 = 2\|\psi\|^2 + 2\|\phi\|^2$.

Proof. Quadraticity on 2D planes implies the parallelogram identity. $\square$

Theorem 4 (Jordan–von Neumann). *There exists a unique inner product inducing $\|\cdot\|$, given by polarization.*

Proof. Standard polarization proof using Theorem 3. $\square$

3 Dynamics Layer: Linearity, Unitarity, Generator, Schrödinger Form

3.1 Linearity from No Carriers

Theorem 5. *Any admissible reversible continuous update law acts linearly on representatives.*

Proof. Nonlinearity makes evolution depend on decomposition; under composition this becomes contextual transport, violating No Carriers. $\square$

3.2 Unitarity from Standing Conservation

Theorem 6. *Standing Conservation implies norm preservation; combined with linearity, updates are unitary.*

Proof. Norm change would change standing weight, violating No Generators. $\square$

3.3 Generator construction

Definition 9. *For a strongly continuous unitary group $\{T_t\}$ define*

$$A\psi = \lim_{t \rightarrow 0} \frac{T_t\psi - \psi}{t}.$$

Theorem 7. *A is skew-adjoint and generates $\{T_t\}$; $H = iA$ is self-adjoint.*

Proof. Use the Laplace-resolvent construction and skew-adjoint Hille–Yosida conditions (as in maximal hardening v6). $\square$

3.4 Schrödinger form

Theorem 8. *For $\psi(t) = T_t\psi_0$ with $\psi_0 \in \mathcal{D}(H)$,*

$$\frac{d}{dt}\psi(t) = -iH\psi(t).$$

Proof. Differentiate $T_t = e^{-itH}$ strongly on $\mathcal{D}(H)$. $\square$

4 Impossibility Gate Denial Modes

- Nonlinearity $\Rightarrow$ Carrier mode.
- Nonunitarity $\Rightarrow$ Generator mode.
- Collapse/noise without witness $\Rightarrow$ Generator + Repair mode.
- Free parameter $\Rightarrow$ Parameter-Illusion mode.

5 Conclusion

Admissibility $\Rightarrow$ Hilbert-or-Denial $\Rightarrow$ Unitary Group $\Rightarrow$ Schrödinger Form.
